

Activity- 4

AIM:

To identify a diode, an LED, a resistor, and a capacitor from a mixed collection of such items.

APPARATUS

Multimeter, a diode, an LED, a resistor, and a capacitor

THEORY:

For identification, the appearance and working of each item will have to be considered.

- i. A diode is a two-terminal device. It conducts when forward-biased and does not conduct when reverse-biased. It does not emit light while conducting. Hence, it does not glow.
- ii. An LED (light-emitting diode) is also a two-terminal device. It also conducts when forward-biased and does not conduct when reverse-biased. It emits light while conducting. Hence, it glows.
- iii. A resistor is a two-terminal device. It conducts when either forward-biased or reverse-biased. (In fact, there is no forward or reverse bias for a resistor.) It conducts even when operated with A.C voltage.
- iv. A capacitor is also a two-terminal device. It does not conduct when either forward-biased or reverse- biased. When a capacitor is connected to a D.C source, the multimeter shows full-scale current initially, but it decays to 0 quickly. It is because initially, a capacitor draws a charge.

PROCEDURE:

1. Check the physical appearance of the components. If the item has two terminals, it may be a diode, an LED, a resistor, or a capacitor.
2. To differentiate, proceed as follows:
 - i. Look for colour bands. If it has a typical set of three colour bands followed by a silver or gold band, the component is a resistor.
 - ii. Connect the multimeter terminals (in resistance mode, highest range) to the component terminals and watch for the multimeter deflection. Also, repeat by reversing the component terminals.
 - iii. If the multimeter shows an equal deflection in both directions, the component is a resistor.
 - iv. If the deflection is accompanied by emission of light, in one direction and a much less or 0 deflection in the other direction, the component is an LED.

- v. If the multimeter does not show any deflection in one direction and shows deflection with no light emission in the other direction, then the component is a diode.
- vi. If the multimeter does not show any deflection on connecting the terminals either way to the component, it is a capacitor, but if the capacitance of the capacitor is large, the multimeter shows a momentary deflection.

RESULT:

A diode, an LED, a resistor, and a capacitor are identified, respectively, from a mixed collection.

Precautions and sources of error

1. All are very delicate electronic instruments. So, careful handling should be followed.
2. While checking the resistance of a component, avoid touching either of the metal ends of the multimeter leads. Body resistance in parallel with component resistance can affect the resistance measurement.

Diagram and observation table on the left side of the page

Diagram:



